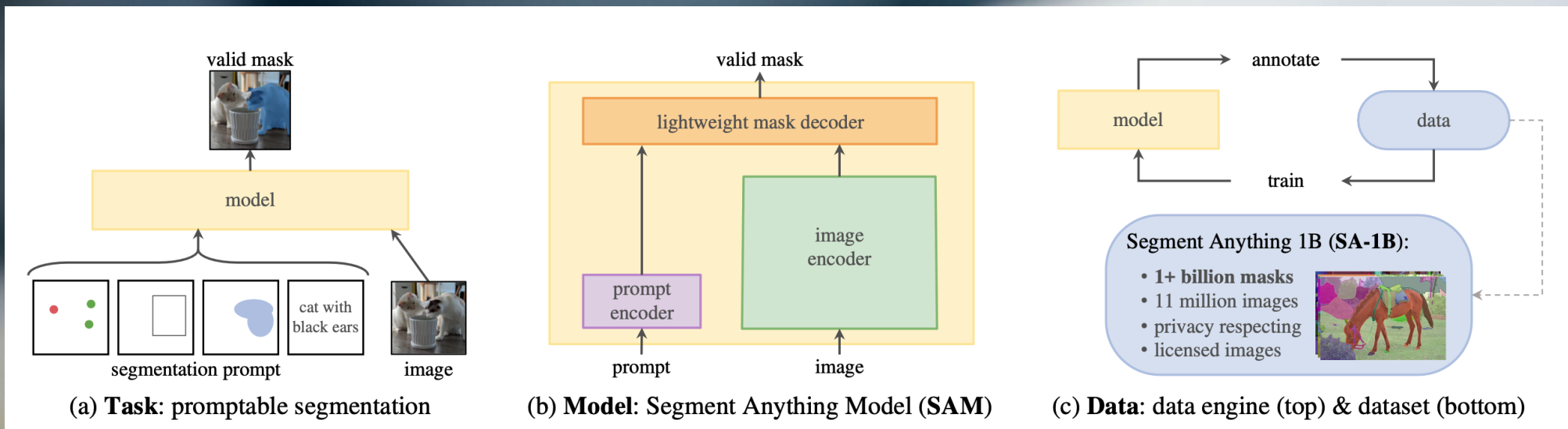


Segment Anything Model (SAM)

Oct 22, 2024

Gist



Runner-up Best Paper ICCV 2023

Goals



PROMPTABLE SEGMENTATION
TASK



PROMPTABLE SEGMENTATION
MODEL



LARGE SCALE TRAINING
DATASET (PROMPTS + MASKS)



Datasets

- SA-1B
- 11M images
- 1B masks



Prompts and Model

- Flexible prompts
 - Points, boxes, masks
- Ambiguity-aware
 - A point can be a subpart, part or whole
- Fast
 - ~50ms – prompt and mask decoder, given image encoding

Data Engine

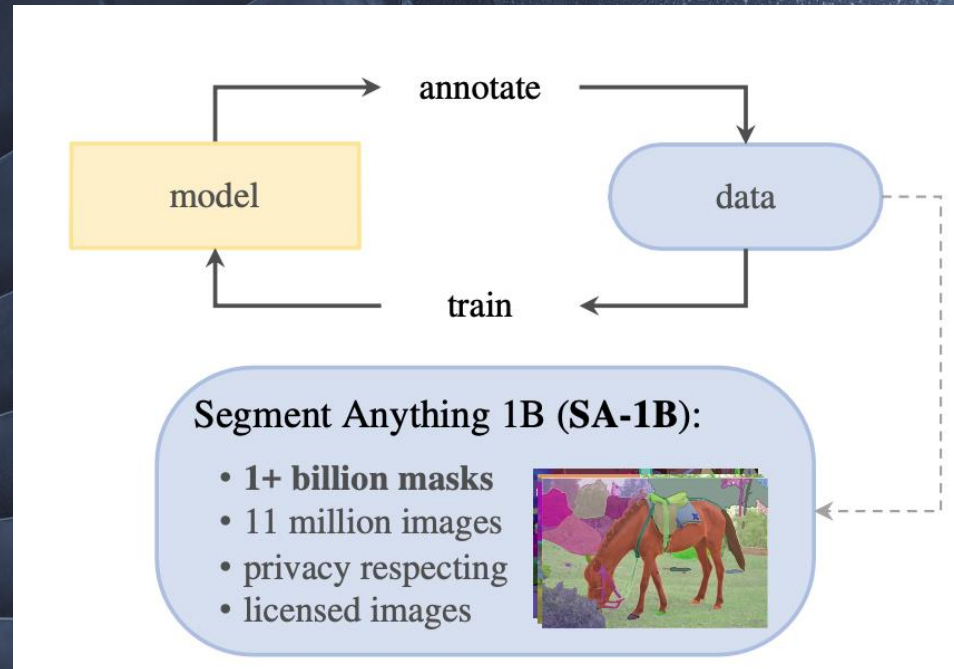


Figure 3: Each column shows 3 valid masks generated by SAM from a single ambiguous point prompt (green circle).

- Start with an initial model trained on existing segmentation dataset
- Assisted-manual: user click fg/bg point, model give a segmap interactively, model retrained with only newly annotated data (6 times)
- Semi-automated: focus on DIVERSITY of masks. Start with high confidence masks, then direct annotators to annotate additional unannotated objects (5 times of retraining)
- Fully automatic: use predicted masks -> select stable masks -> NMS

- Image Encoder – MAE ViT
- Prompt Encoder – Position encodings for points/boxes, CLIP for text
- Mask decoder – transformer decoder block -> prompt<->image self+cross attn -> upsample -> probability at each position in mask
- Predict multiple masks ranked in terms of confidence scores (estimated IOU)

Focal Loss:

$$FL(p_t) = -(1 - p_t)^\gamma \log(p_t)$$

Eq. 6. Non-alpha form

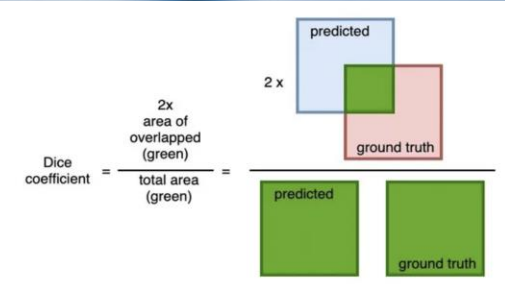
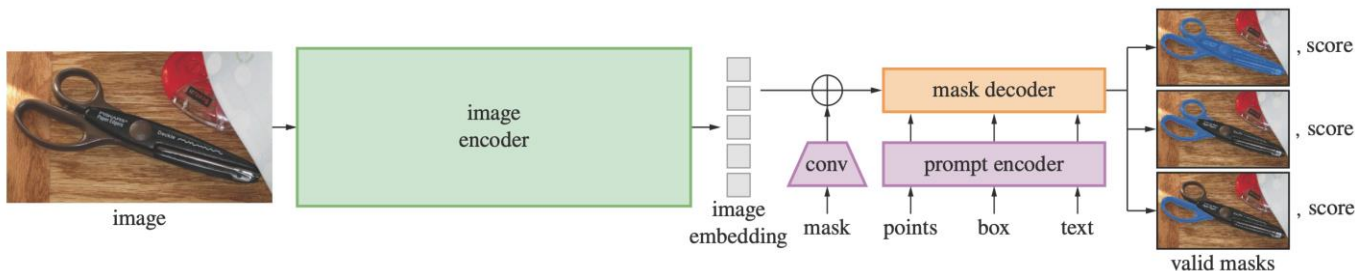
$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

Eq. 7. Alpha Form

DICE Loss:

$$DiceLoss(y, \bar{p}) = 1 - (2y\bar{p} + 1) / (y + \bar{p} + 1)$$

Training/Losses



Data Quality

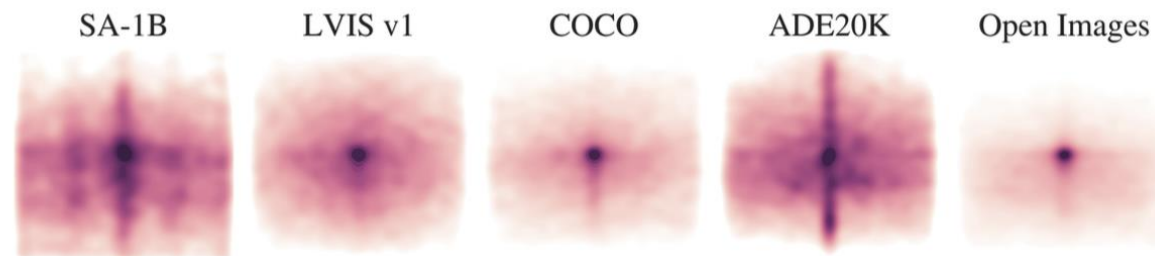


Figure 5: Image-size normalized mask center distributions.

Mask quality. To estimate mask quality, we randomly sampled 500 images (~50k masks) and asked our professional annotators to improve the quality of all masks in these images. Annotators did so using our model and pixel-precise “brush” and “eraser” editing tools. This procedure resulted in pairs of automatically predicted and professionally corrected masks. We computed IoU between each pair and found that 94% of pairs have greater than 90% IoU (and 97% of pairs have greater than 75% IoU). For comparison,



Attendance

SAM2 – Images and VIDEOS

INTRODUCING

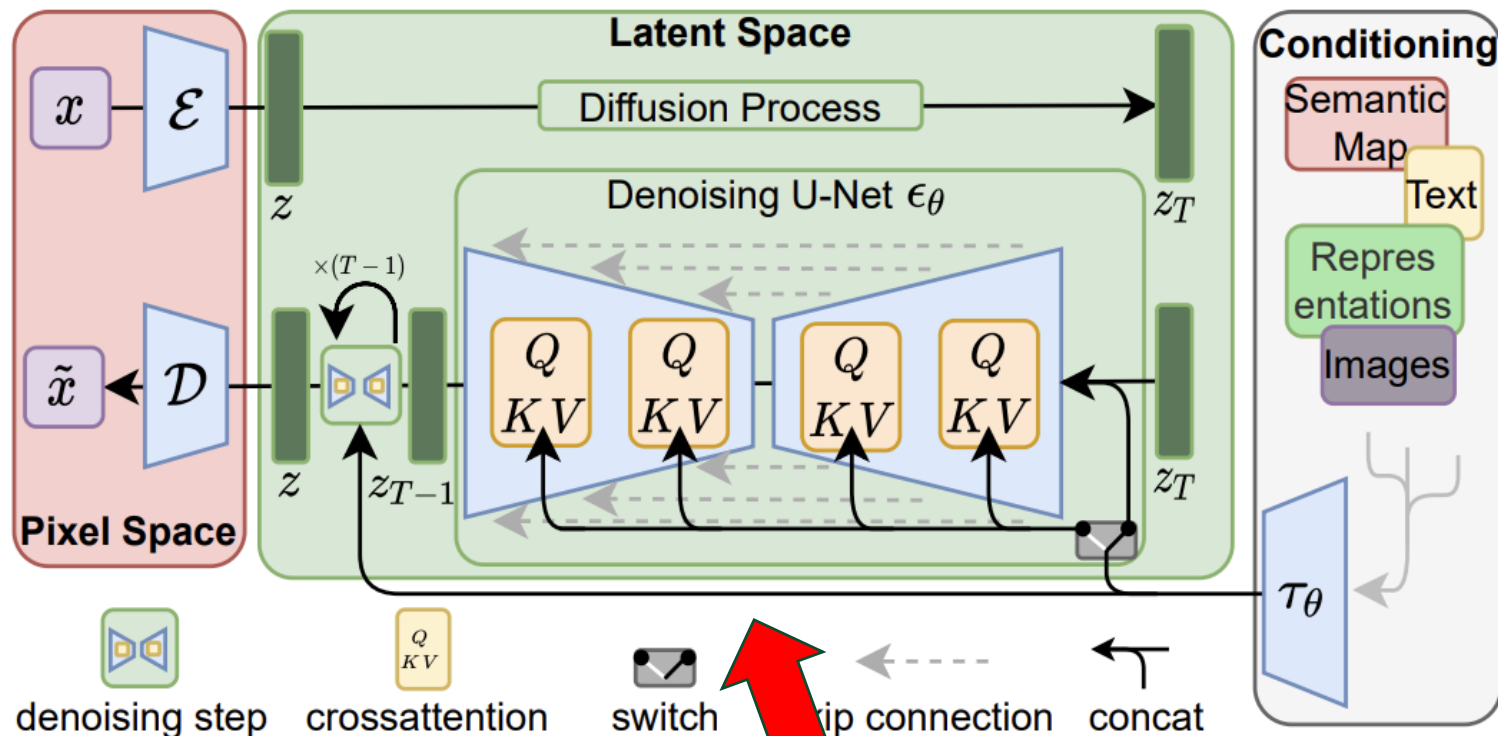
**Meta Segment
Anything Model 2
for videos and images**



 Meta

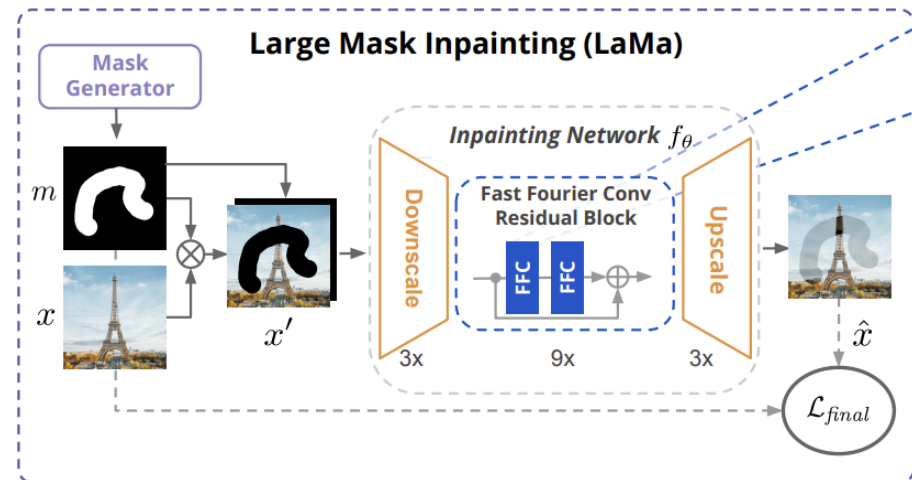
AI at Meta

SDXL Recap



1000 steps (DDIM vs DDPM)

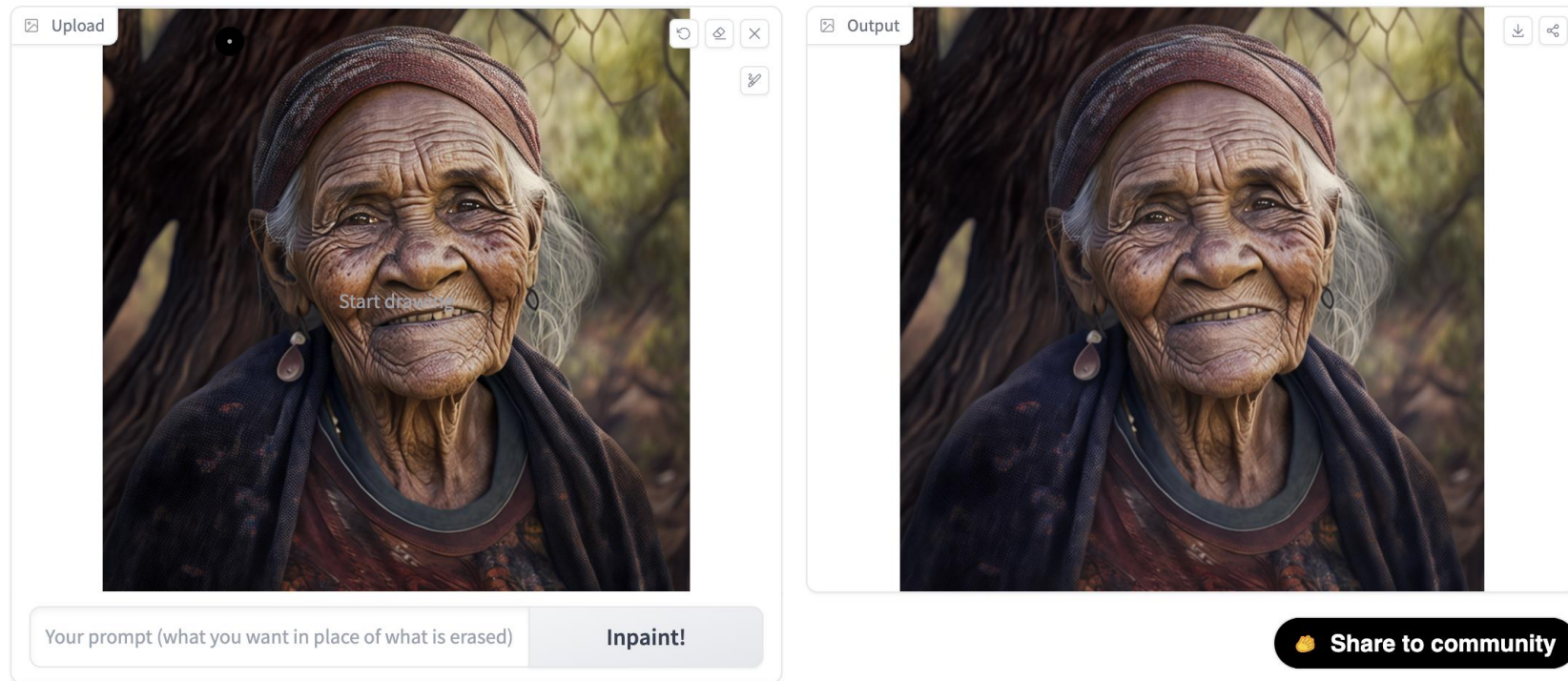
How do they train SD inpainting?



SDXL Inpainting to Remove Object

Stable Diffusion XL Inpainting 🎨

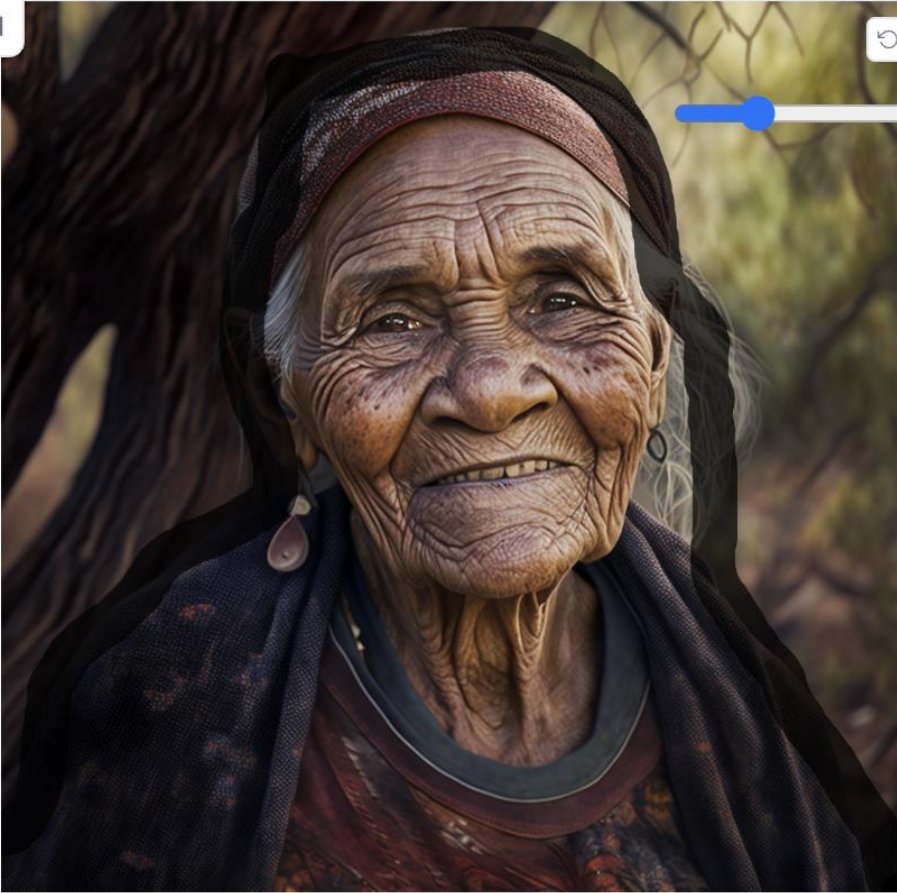
Demo for the [Stable Diffusion XL Inpainting model](#), add a mask and text prompt for what you want to replace

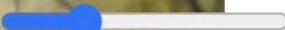


Stable Diffusion XL Inpainting 🎨

Demo for the [Stable Diffusion XL Inpainting model](#), add a mask and text prompt for what you want to replace

 Upload





Your prompt (what you want in place of what is erased)

Inpaint!

 Output

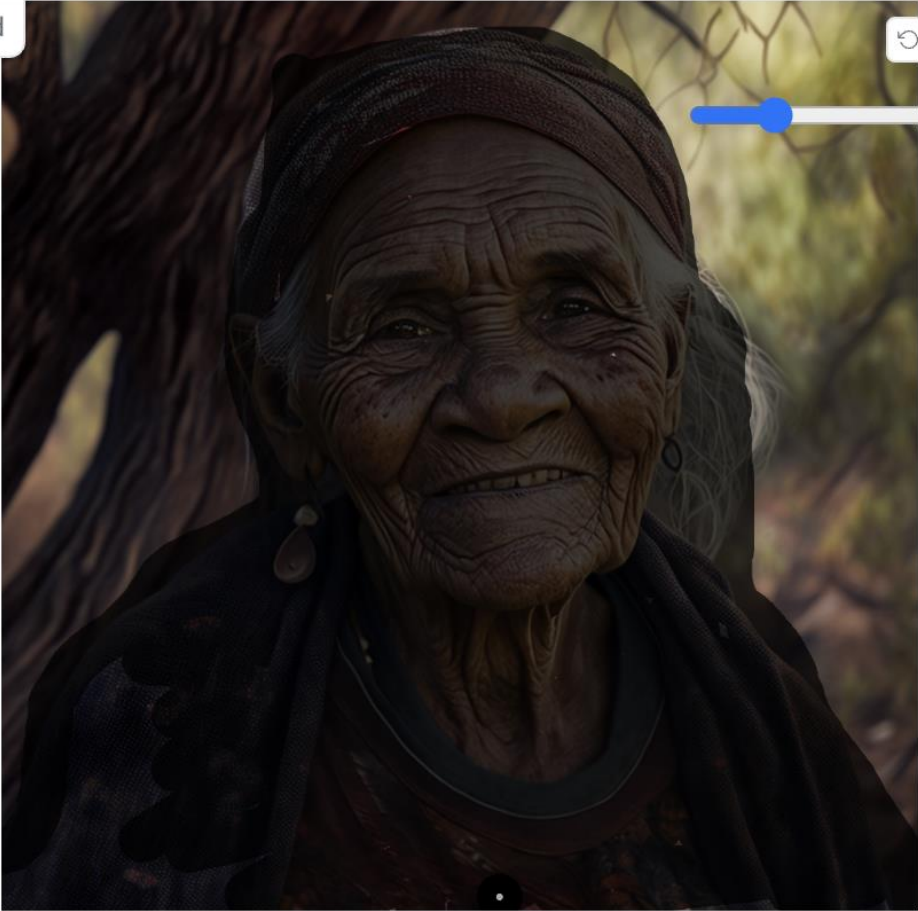


 Share to community

Stable Diffusion XL Inpainting 🎨

Demo for the [Stable Diffusion XL Inpainting model](#), add a mask and text prompt for what you want to replace

 Upload





Your prompt (what you want in place of what is erased)

Inpaint!

 Output

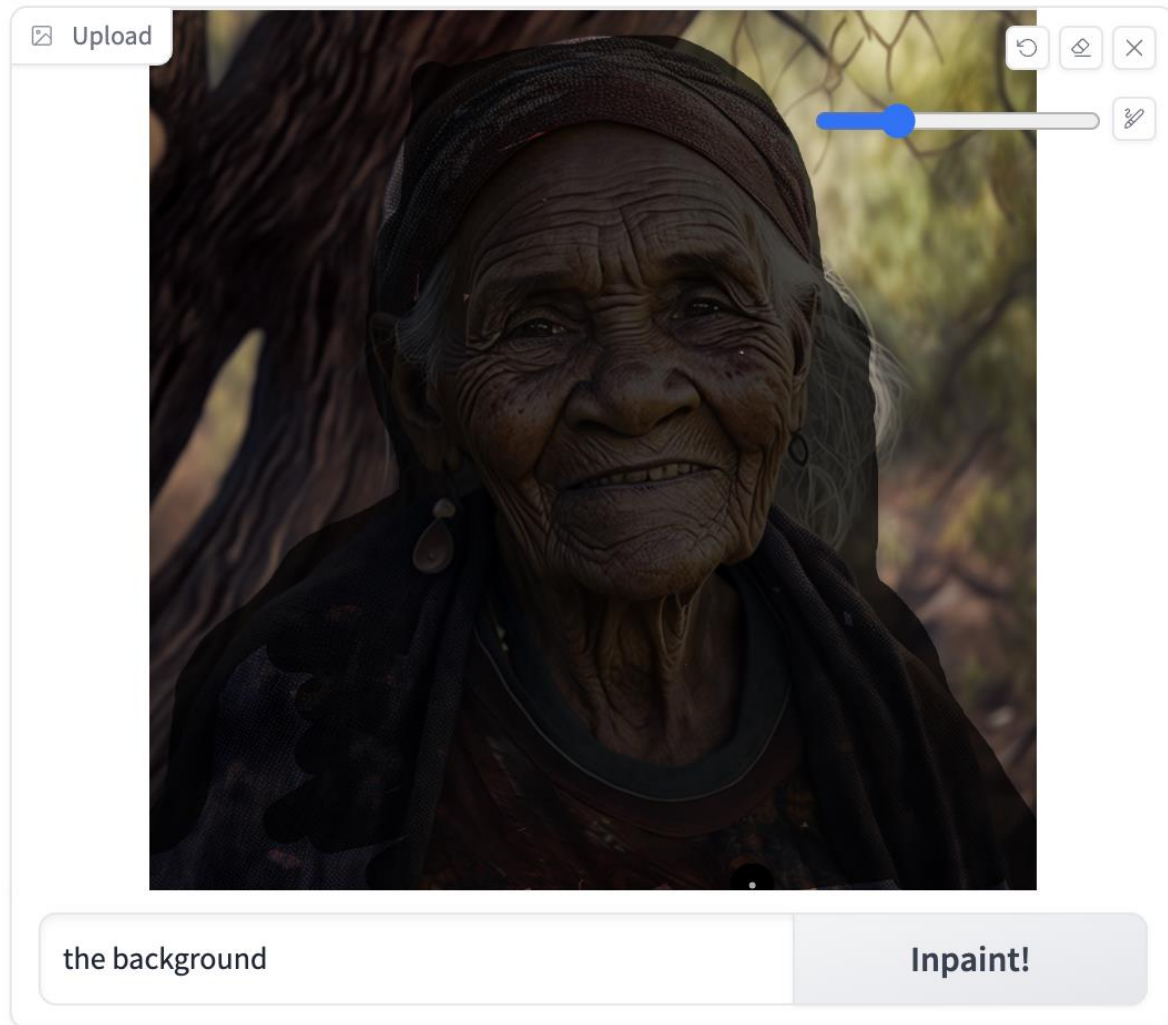




 **Share to community**

Stable Diffusion XL Inpainting 🎨

Demo for the [Stable Diffusion XL Inpainting model](#), add a mask and text prompt for what you want to replace

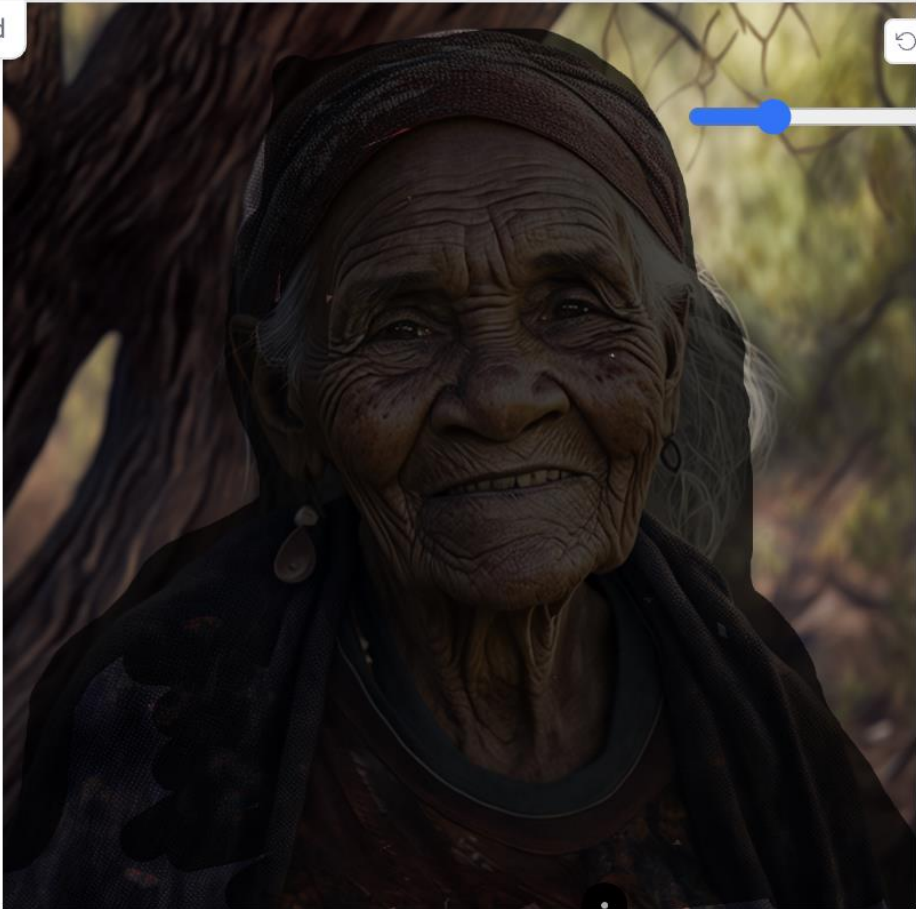


👉 Share to community

Stable Diffusion XL Inpainting 🎨

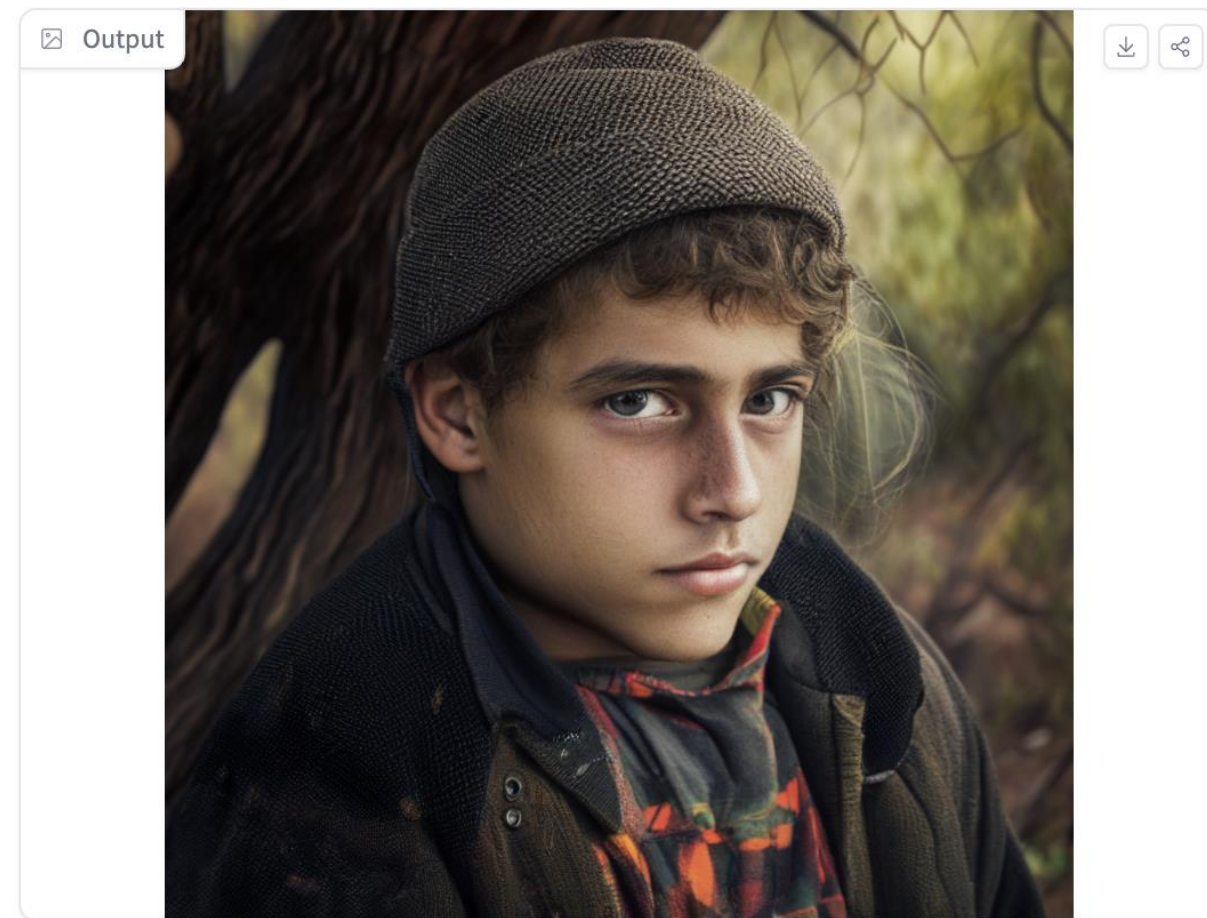
Demo for the [Stable Diffusion XL Inpainting model](#), add a mask and text prompt for what you want to replace

Upload



a young man

Inpaint!

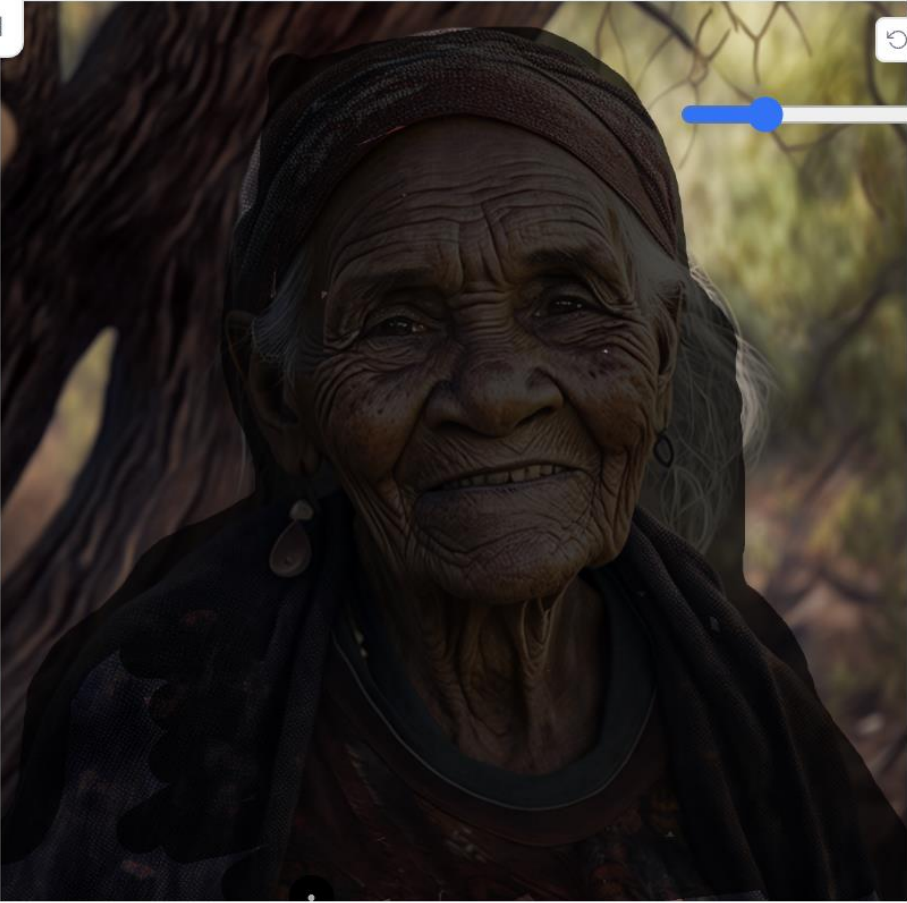


 **Share to community**

Stable Diffusion XL Inpainting 🎨

Demo for the [Stable Diffusion XL Inpainting model](#), add a mask and text prompt for what you want to replace

Upload



↶

🗑️

✕

🖌️

empty

Inpaint!

Output



⬇️

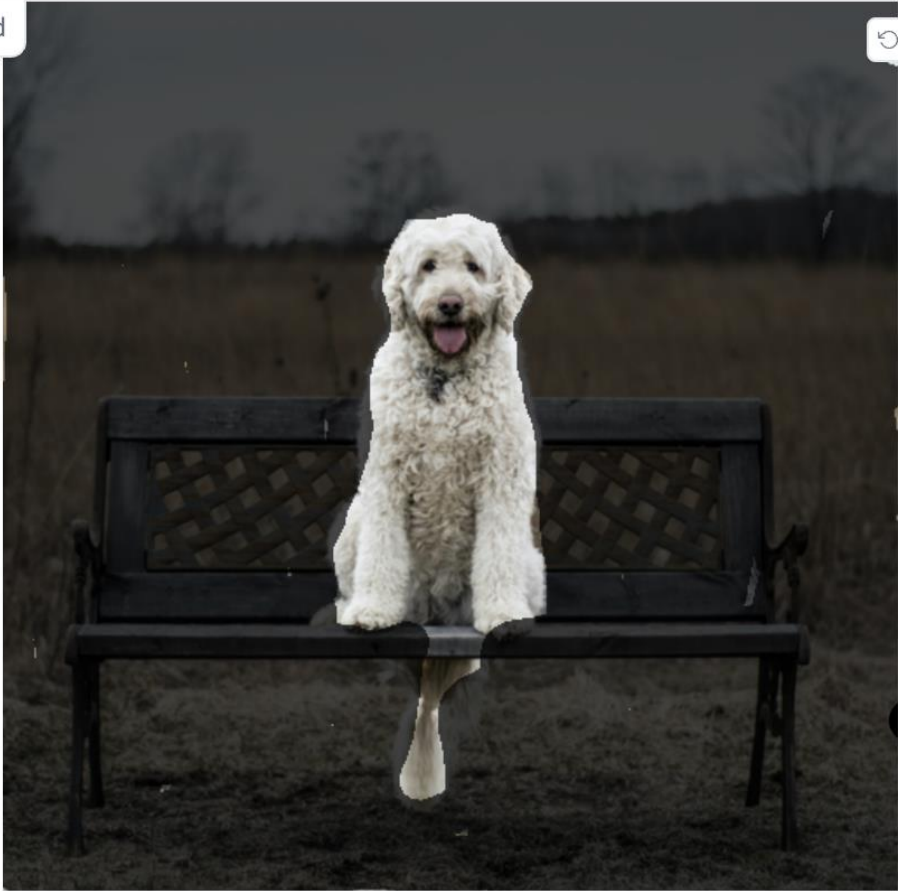
🔗

👉 Share to community

Stable Diffusion XL Inpainting 🎨

Demo for the [Stable Diffusion XL Inpainting model](#), add a mask and text prompt for what you want to replace

 Upload





Your prompt (what you want in place of what is erased)

Inpaint!

 Output



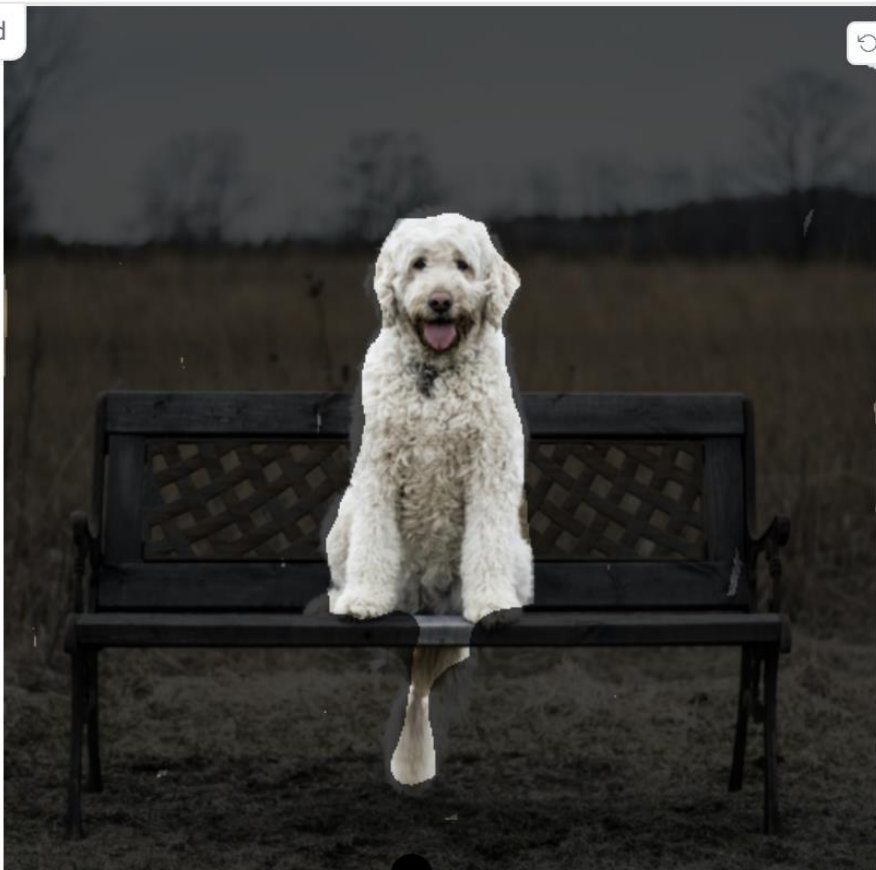


 Share to community

Stable Diffusion XL Inpainting 🎨

Demo for the [Stable Diffusion XL Inpainting model](#), add a mask and text prompt for what you want to replace

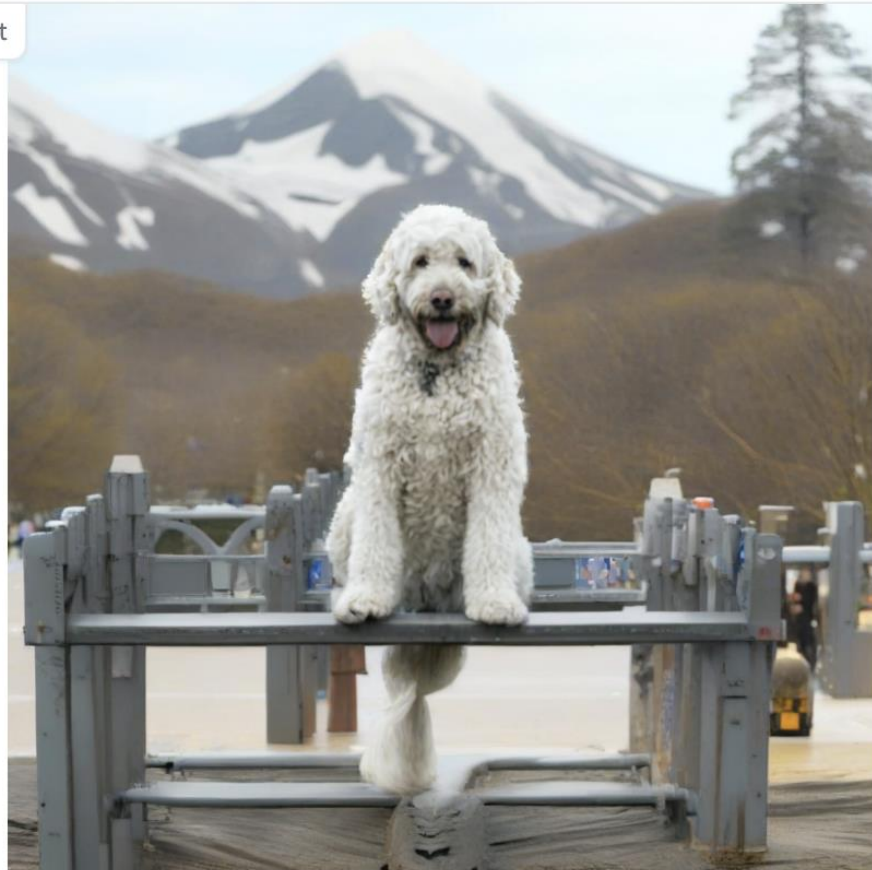
Upload



snow mountain

Inpaint!

Output



 Share to community

Stable Diffusion XL Inpainting 🎨

Demo for the [Stable Diffusion XL Inpainting model](#), add a mask and text prompt for what you want to replace

 Upload



a giraffe

Inpaint!

 Output



 Share to community

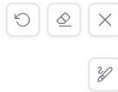
SD-1.5 Inpainting Works!!

Stable-Inpainting 1.5

☐ Use RASG

Inpainting Prompt

Input Image



Assignment Due 10/29

Take cityscapes dataset:

- Goal1 -- Inpainting: Delete all the vehicles in the images
 - Meaning: inpainting to remove all the cars, trucks etc in the images
- Goal2 -- outpainting: Delete all the background except the vehicles
 - Meaning: outpainting to remove the background and replace with SNOW background (use the text prompt)
- Steps:
 - Detect cars, trucks, etc. (suggestion: open-world detection with grounding-DiNO)
 - Run SAM on the boxes to get mask
 - Run SD-1.5-Inpainting on the mask